

# PAGADALA CHARAN KARTHIK

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Rly Kodur, Andhra Pradesh, India — 2027 Graduate

## Profile Summary

Computer Science Engineering student (CGPA: 9.37/10) with strong foundations in Data Structures, Algorithms, Object-Oriented Design, and distributed systems. Hands-on experience building scalable production-grade software in Python, Java, C, and JavaScript. Two technical internships at Microsoft Azure and Rinex – proven ability to design, build, and deliver high-quality software in agile environments at scale.

## Education

### Mohan Babu University

B.Tech in Computer Science Engineering | CGPA: 9.37 / 10 | Expected Graduation: 2027

Tirupati, Andhra Pradesh

2023 – 2027

### Narayana Junior College

Intermediate (Class XII) – BIEAP | 92%

Andhra Pradesh

2021 – 2023

### Sri Gouri Sankar Vidyalayam

Secondary School (Class X) – BSEAP | 99%

Andhra Pradesh

2016 – 2021

## Technical Skills

**Languages:** Python, Java, C, JavaScript, TypeScript, SQL

**Data Structures & Algorithms:** Arrays, Strings, Linked Lists, Trees, Graphs, Sorting, Searching, Dynamic Programming, Recursion

**Core CS:** Object-Oriented Design, Operating Systems, DBMS, Computer Networks, Complexity Analysis, System Design

**Distributed Systems:** Scalable REST APIs, Microservices, Docker, Redis (caching), WebSockets, Fault-Tolerant Architecture

**Web & Frameworks:** React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS

**AI / ML:** PyTorch, Scikit-learn, LangChain, FAISS, OpenAI API, NLP, Model Evaluation

**Databases:** MongoDB, Supabase (PostgreSQL), MySQL, Firebase, Redis, Neo4j

**Cloud & Tools:** AWS (basics), Microsoft Azure (AZ-900), Google Cloud, Docker, Git, GitHub, Postman, Agile/Scrum

## Work Experience

### Cloud & AI Intern

Jan 2026 – Feb 2026

Microsoft Azure

- Designed and deployed scalable cloud infrastructure on Azure including App Service, Functions, and Blob Storage – gained hands-on experience building fault-tolerant, production-grade distributed services.

### AI & Machine Learning Intern

Jan 2025 – Mar 2025

Rinex

- Built and evaluated ML models on real-world datasets using Python – applied feature engineering, hyperparameter tuning, and complexity analysis to deliver optimised, scalable solutions.

## Projects

### RoofVision AI | Python, PyTorch, Mask R-CNN, FastAPI, React, Docker | View

- Built a scalable computer vision system verifying rooftop solar installations via satellite imagery – designed for 10,000 claims/day via Docker Compose with FastAPI inference at 98% lower cost than manual review.
- Applied OOP design and distributed service architecture to deliver a fault-tolerant, production-grade system – directly demonstrates ability to design and code solutions from broadly defined problems at Amazon scale.

### AI Knowledge Graph Builder | Python, Neo4j, FAISS, Groq Llama-3, LangChain, Streamlit | View

- Architected a distributed hybrid retrieval system combining FAISS vector search and Neo4j graph traversal for scalable Q&A over 500K+ Enron emails – demonstrates distributed storage, index, and query system design.
- Led team delivery across 4 Agile milestones from schema to Streamlit Cloud deployment – shipped a live product on time while collaborating with cross-disciplinary team members.

### TeamFlow Pro | Next.js 15, Node.js, Express.js, MongoDB, JWT, Docker, Railway | View

- Built an enterprise task management platform with RBAC, Kanban board, and live analytics dashboard – engineered scalable REST APIs with JWT authentication and Mongoose ODM, deployed on Railway.

## Certifications & Achievements

- **Microsoft Azure Fundamentals (AZ-900)** — Cloud architecture, distributed systems, Azure AI services, security & networking
- **Infosys Generative AI Certification** — Prompt engineering, LLM applications, generative AI
- **Google Cloud Arcade** — AI/ML skill labs, data pipelines, and multiple Cloud skill badges
- **NIAT Gaming Hackathon 2025** — Top 8 nationally out of 100+ participants
- **Competitive Programming** — Active on LeetCode & HackerRank (DSA, Algorithms, Problem Solving)